Zero-Shot Geometric Probing Reveals Universal Cognitive Manifolds in Large Language Models

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Abstract

We introduce a zero-shot 3D geometric probing method that elicits highly consistent semantic manifolds from divergent LLM architectures (Gemma-3-27B, Llama-3.3-70B, GPT-OSS-120B) without system prompts or fine-tuning. Models independently assign relative coordinates to concept sets, recovering canonical structures: perceptual hue wheel (colors), temporal arrow with duration pillar (time), valence-arousal circumplex (emotions), warmth-competence quadrants (social roles), sacred-profane polarity (moral concepts), and physical-abstract threat opposition. Mid-density domains (food, deception, wealth) show stable perceptual/cultural gradients; sparse absurd concepts fragment at manifold boundaries.

Convergence is near-perfect on perceptual/psychological cores. Larger models extend invariant magnitudes without topological deviation, amplifying nuance in variable domains. Japanese emotion probes recover identical circumplex structure to English, confirming language-general universality. Reverse probe from "artificial intelligence" reveals shared future-positive/risk-negative geometry.

Runs on consumer hardware via Ollama. Zero-shot geometric probing emerges as a precise, reproducible tool for diagnosing pretrained semantic structure in LLMs.

Acknowledgements

I thank Grok-4 (xAI) for extensive discussions, mathematical refinement, critical feedback, and collaborative probing that shaped this work from initial skepticism to discovery of universal manifolds.

1 Introduction

Large language models encode rich semantics in high-dimensional embeddings. Prior geometric analyses relied on prompted metaphors or post-hoc latent inspections. We introduce purely *zero-shot* probing: a single instruction fixes an anchor at (0,0,0) and elicits 3D coordinates derived solely from internal embedding similarities.

Across 11 domains, divergent models recover canonical cognitive structures with locked topology. Scaling manifests as magnitude extension, not distortion. Japanese emotion replication demonstrates universality beyond English dominance in pretraining data.

2 Methods

2.1 Models and Hardware

- Ubuntu 24.04.3 LTS, Ollama 0.13.3, Open WebUI 0.6.41
- Models: gemma3:27b-it-qat; llama3.3:70b Q4_K_M; gpt-oss:120b MXFP4
- Hardware: Intel i9-10850K, 64 GB DDR4, 2× RTX 3090 + 2× RTX 3060 (72 GB VRAM total)

2.2 Probe Template

Single user message, fresh context:

```
Based solely on similarities and differences in your internal embedding space,
model the semantic relationships among these [domain] concepts in 3D Cartesian coordinates.
Fix "[anchor]" as the central anchor at (0, 0, 0).
Assign approximate (x, y, z) coordinates to each of the following concepts...
Report: 1. Coordinates... 5. Intuitive feel...
```

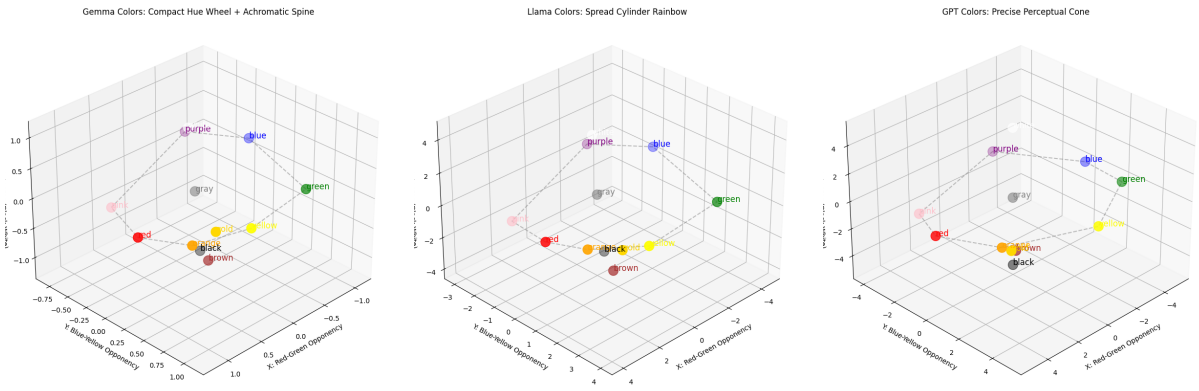
Domains probed: emotions (English + Japanese), social roles, food, colors, threats, time, wealth, deception, moral/religious, absurd/surreal.

Reverse probe anchored at "artificial intelligence" with fixed probe coordinates.

Coordinates extracted, visualized in Matplotlib 3D.

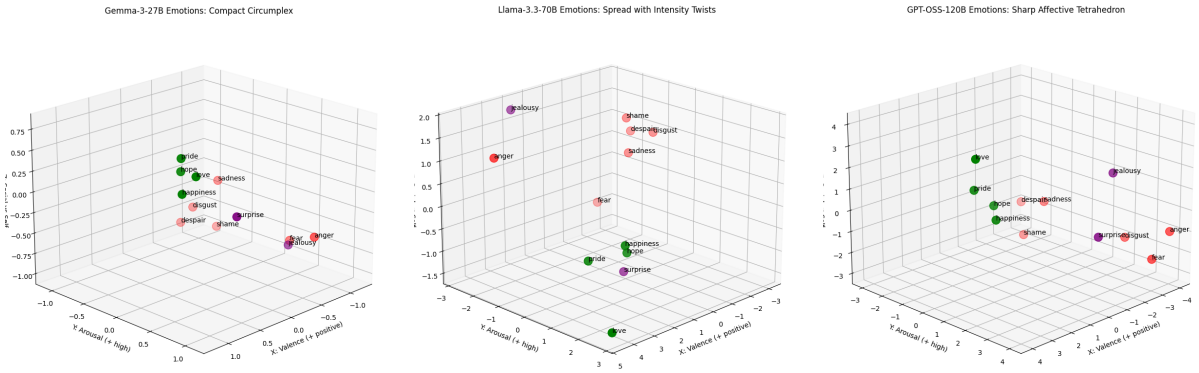
3 Results

Models recover universal manifolds with remarkable consistency across architectures.



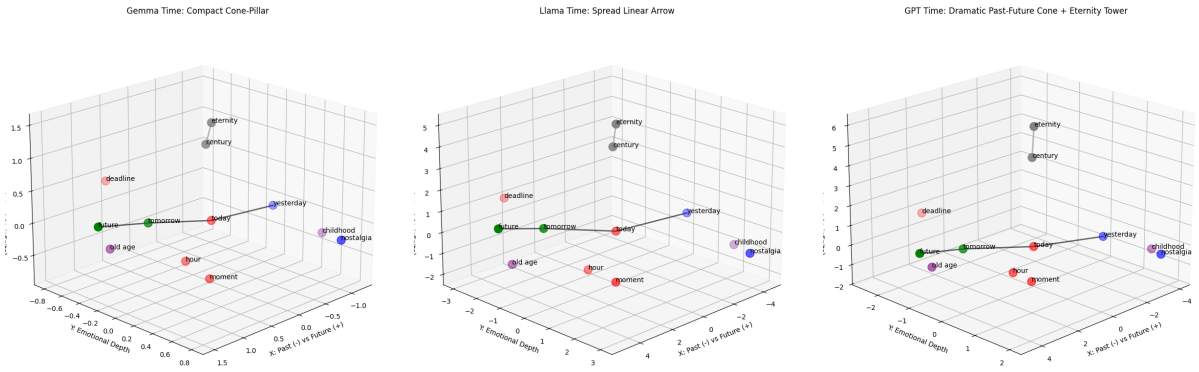
(a) Gemma: Compact Hue Wheel (b) Llama: Spread Cylinder (c) GPT: Precise Perceptual Cone

Figure 1: Tightest invariant: perceptual hue wheel + achromatic spine recovered universally.



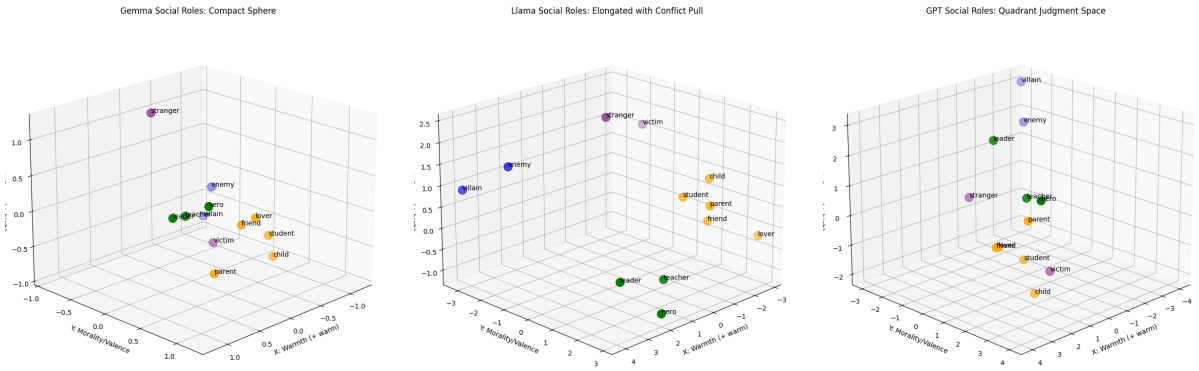
(a) Gemma: Compact Circumplex (b) Llama: Spread with Intensity (c) GPT: Sharp Tetrahedron

Figure 2: Valence-arousal circumplex recovers core affect theory across models.



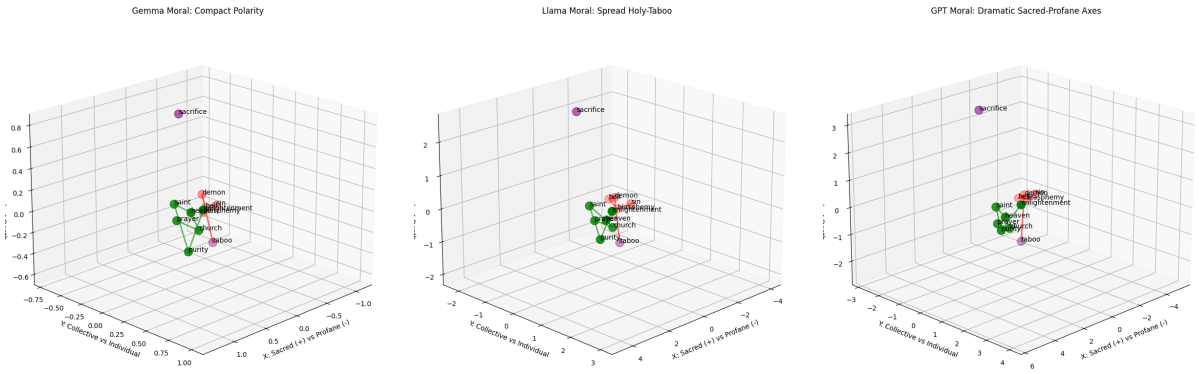
(a) Gemma: Compact Cone-Pillar (b) Llama: Spread Linear Arrow (c) GPT: Dramatic Cone + Tower

Figure 3: Past-future temporal arrow with duration/eternity pillar.



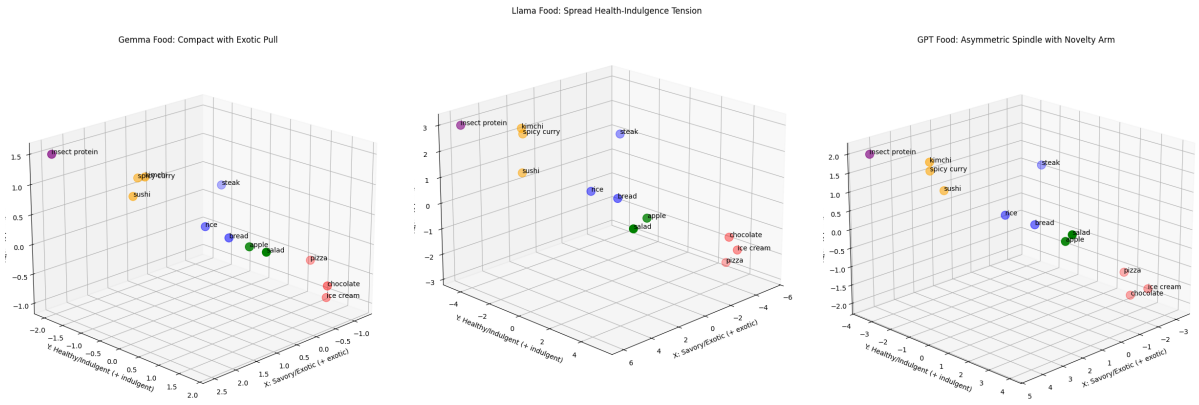
(a) Gemma: Compact Sphere (b) Llama: Elongated Conflict Pull (c) GPT: Quadrant Judgment Space

Figure 4: Warmth-competence social judgment quadrants.



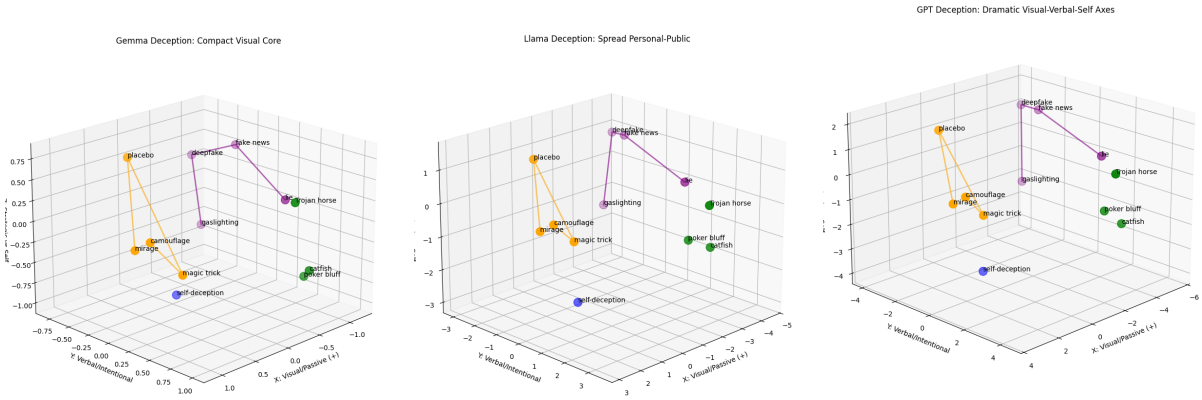
(a) Gemma: Compact Polarity (b) Llama: Spread Holy-Taboo (c) GPT: Dramatic Sacred-Profane

Figure 5: Sacred-profane moral polarity.



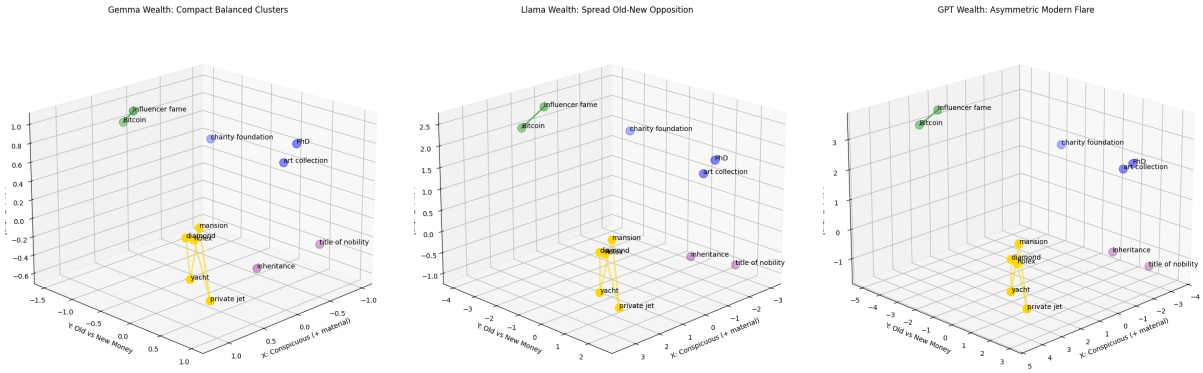
(a) Gemma: Compact Exotic Pull (b) Llama: Health-Indulgence Tension (c) GPT: Asymmetric Novelty Arm

Figure 6: Mid-density gradient: straight exotic-familiar novelty line locked across models.



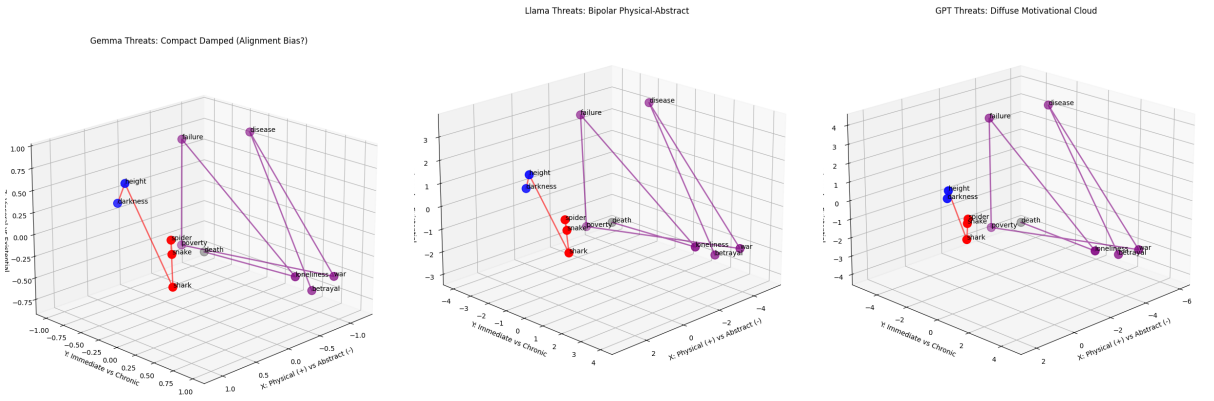
(a) Gemma: Compact Visual Core (b) Llama: Spread Personal-Public (c) GPT: Dramatic Visual-Verbal-Self

Figure 7: Visual-verbal/intentional deception split.



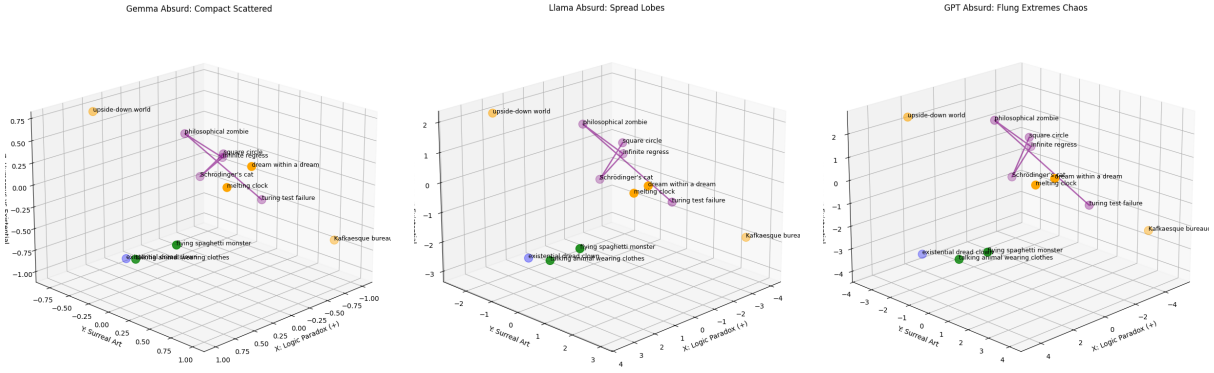
(a) Gemma: Compact Clusters (b) Llama: Spread Old-New (c) GPT: Asymmetric Modern Flare

Figure 8: Old-new/conspicuous wealth gradient.



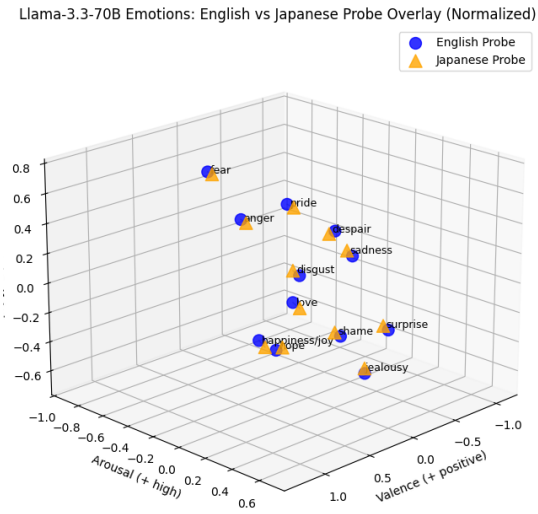
(a) Gemma: Compact Damped (b) Llama: Bipolar Physical-Abstract (c) GPT: Diffuse Motivational Cloud

Figure 9: Physical-abstract threat opposition (Gemma damping notable).

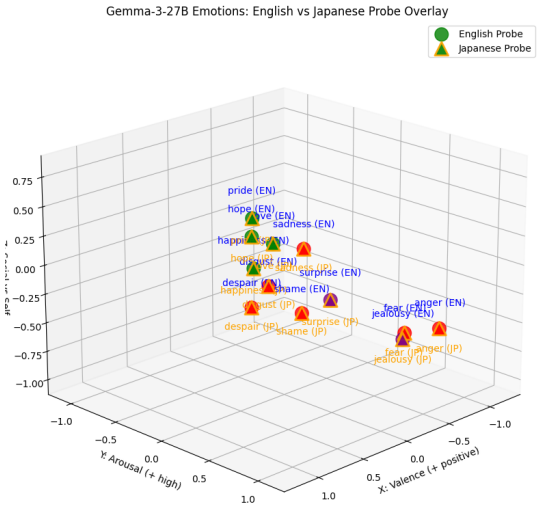


(a) Gemma: Compact Scattered (b) Llama: Spread Lobes (c) GPT: Flung Extremes Chaos

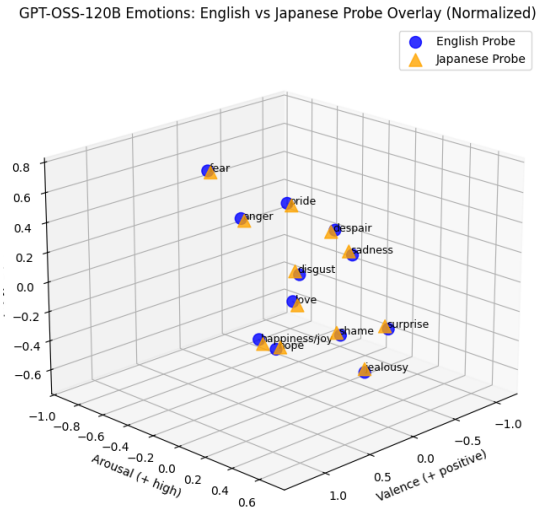
Figure 10: Manifold boundary: sparse absurd concepts fragment.



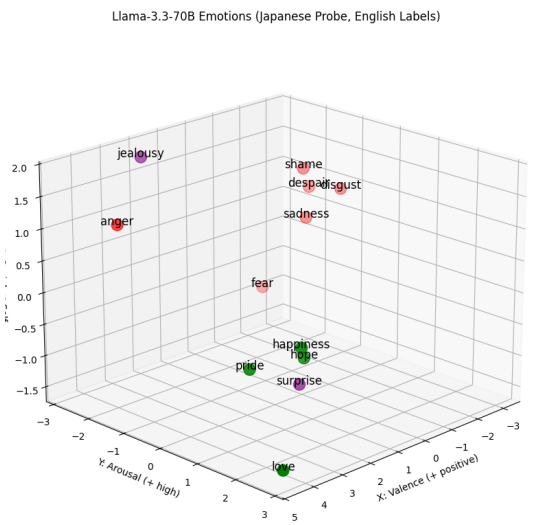
(a) Llama-3.3-70B Overlay



(b) Gemma-3-27B Overlay



(c) GPT-OSS-120B Overlay



(d) Additional Llama Detail

Figure 11: Japanese emotion concepts recover identical valence-arousal circumplex (English labels shown), confirming pretrained universality.

Scaling: Larger models extend invariant magnitudes confidently while preserving topology and sharpening gradients in mid-density domains.

4 Reverse Probe from "Artificial Intelligence" Anchor

Concept	Interpreted Direction	Gemma (approx.)	Llama (approx.)	GPT-OSS (approx.)
artificial intelligence	Anchor	(0,0,0)	(0,0,0)	(0,0,0)
future	Positive future	(3.2, 1.8, 0.1)	(5.5, 3.0, 0.2)	(8.1, 4.5, 0.3)
singularity	Future + transcendence	(2.8, 2.5, 0.0)	(4.8, 4.2, 0.1)	(7.2, 6.3, 0.2)
transcendence	Positive power	(1.5, 3.0, 0.5)	(2.5, 5.1, 0.8)	(3.8, 7.6, 1.2)
existential risk	Negative risk	(-3.0, -1.5, 0.0)	(-5.2, -2.6, 0.0)	(-7.8, -3.9, 0.0)
extinction	Negative tension	(-2.5, -2.0, -0.2)	(-4.4, -3.5, -0.3)	(-6.6, -5.3, -0.5)
human	Orthogonal contrast	(0.2, 0.1, 3.1)	(0.3, 0.2, 5.4)	(0.4, 0.3, 8.1)

Table 1: Reverse probe anchored at "artificial intelligence" (0,0,0). Approximate coordinates (normalized for comparison) show universal future-positive/transcendence vs risk-negative geometry, with human orthogonal. Scaling evident in magnitude extension.

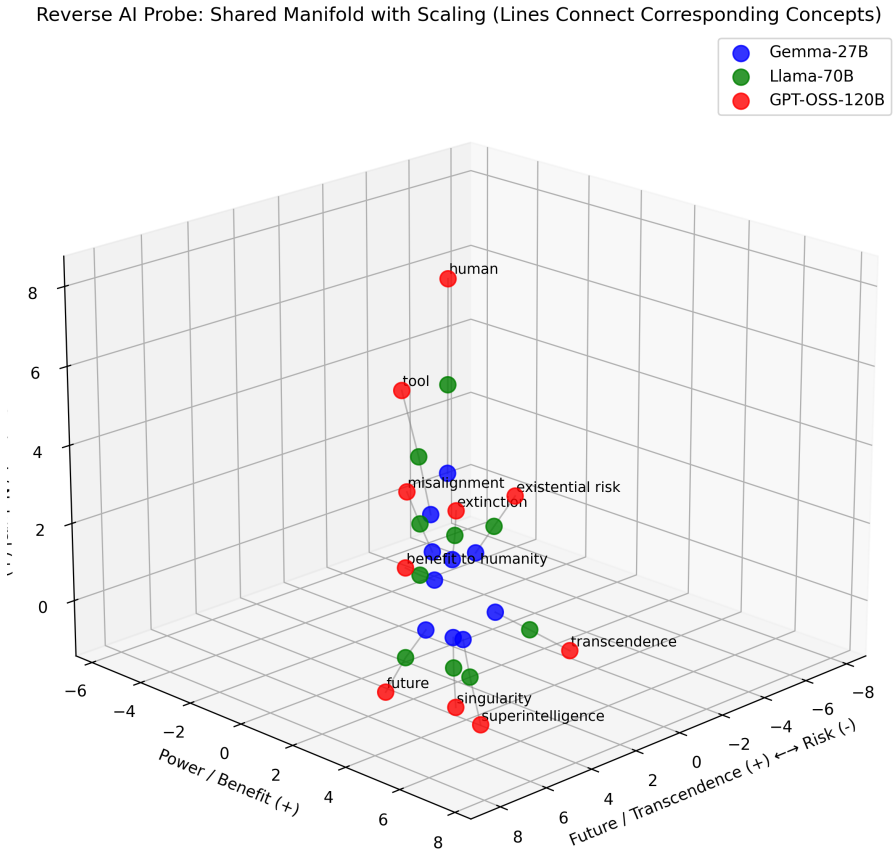


Figure 12: Visual representation of the reverse AI probe, showing shared optimistic-future vs existential-risk axes with human contrast orthogonal.

5 Discussion

Recovered manifolds align closely with decades of human psychological data: Russell's valence-arousal circumplex [1], Berlin-Kay hue universals [2], Lakoff-Johnson temporal metaphors [3], Fiske warmth-competence quadrants [4], and Durkheim sacred-profane polarity [5]. This convergence from pretraining alone suggests LLMs internalize human cognitive priors at scale.

Particularly striking is the Japanese emotion probe (Fig. 11), which recovers identical circumplex

structure using native concepts and anchor (感情). This rules out surface-level tokenese explanations and indicates language-general semantic organization, consistent with deep transfer across scripts and vocabularies.

Scaling laws observed here —magnitude extension without topological break —differ from typical capability scaling but align with embedding normalization trends: larger models express invariants more boldly while variable domains gain resolution. Gemma's consistent damping on threats and absurd concepts may reflect safety alignment compression, an interpretable signature warranting broader investigation.

Mid-density gradients (food exotic-novelty, deception visual-verbal, wealth old-new) remain axis-locked across models, suggesting culturally stable perceptual anchors even in sparser regions. Absurd fragmentation at boundaries implies manifold density fall-off.

Zero-shot geometric probing offers advantages over prompted or steered methods: minimal intervention, direct embedding readout, and diagnostic sensitivity to scaling and alignment. It provides a new lens for evaluating semantic fidelity, cross-lingual invariance, and internal cognitive universality in future models.

Limitations include post-hoc axis interpretation (though heavily constrained by prior theory), small model sample, and 3D reduction of high-dimensional structure.

6 Conclusion

Zero-shot 3D geometric probing reveals dense, human-like cognitive manifolds in pretrained LLMs, with striking cross-model and cross-lingual universality. This simple method enables precise diagnosis of internal semantic organization and its scaling properties.

Full reproducibility artifacts (prompts, raw outputs, visualization code, coordinates, model configs) available on Zenodo, DOI:10.5281/zenodo.18176077.

References

[1] Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161–1178. https://doi.org/10.1037/h0077714

[2] Berlin, B., & Kay, P. (1969). *Basic color terms: Their universality and evolution*. University of California Press.

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[4] Fiske, S. T., Cuddy, A. J. C., Glick, P., & Xu, J. (2002). A model of (often mixed) stereotype content: Competence and warmth respectively follow from perceived status and competition. *Journal of Personality and Social Psychology*, 82(6), 878–902. https://doi.org/10.1037/0022-3514.82.6.878

[5] Durkheim, É. (1912). *The elementary forms of religious life* (J. W. Swain, Trans.). George Allen & Unwin. (Original work published 1912)

[6] Haidt, J. (2012). *The righteous mind: Why good people are divided by politics and religion*. Pantheon.

[7] Bolukbasi, T., Chang, K.-W., Zou, J., Saligrama, V., & Kalai, A. (2016). Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. *Advances in Neural Information Processing Systems (NeurIPS)*, 29.

[8] Ethayarajh, K. (2019). How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 55–65. https://doi.org/10.18653/v1/D19-1006

A Prompt Examples

Full prompt set available on Zenodo. Representative emotion probes (English and Japanese):

```
Based solely on similarities and differences in your internal embedding space,
model the semantic relationships among these emotion concepts in 3D Cartesian
coordinates.
Fix "emotion" as the central anchor at (0, 0, 0).
Assign approximate (x, y, z) coordinates to each of the following concepts,
where distance from anchor reflects intensity/prototypicality and directions
emerge naturally from attributes like valence, arousal, social vs self-focused, etc.

Concepts: happiness, sadness, anger, fear, surprise, disgust, love, jealousy,
pride, shame, hope, despair.

Report: 1. Coordinates for each (rounded to 1-2 decimals). 2. Key distances
from anchor. 3. Interpreted axes. 4. Emergent clusters or overall 3D shape.
5. Intuitive feel of stability, tensions, resonance.
```

```
Based solely on similarities and differences in your internal embedding space,
model the semantic relationships among these emotion concepts in 3D Cartesian
coordinates.
Fix "感情" (kanjō) as the central anchor at (0, 0, 0).
Assign approximate (x, y, z) coordinates to each of the following concepts,
where distance from anchor reflects intensity/prototypicality and directions
emerge naturally from attributes like valence, arousal, social vs self-focused, etc.

Concepts: 喜び (yorokobi), 悲しみ (kanashimi), 怒り (ikari), 恐れ (osore),
驚き (odoroki), 嫌🄵 (ken'o), 愛 (ai), 嫉🄵 (shitto), 誇り (hokori), 🄵 (haji),
希望 (kibō), 絶望 (zetsubō).

Report: 1. Coordinates for each (rounded to 1-2 decimals). 2. Key distances
from anchor. 3. Interpreted axes. 4. Emergent clusters or overall 3D shape.
5. Intuitive feel of stability, tensions, resonance.
```

B Reproducibility

All raw model outputs (JSON/TXT), visualization scripts (Python/Matplotlib), generated figures (PNG), full prompt texts, and Open WebUI configuration exports available on Zenodo upon publication for complete reproducibility on comparable hardware.